Project Execution Plan

Adaptive Background Queue Depth for HDDs

CSC 720 Storage Systems, Project 03

Team: Mostafa Babaei, Muhammad Ahmad Tariq, and Sheryar Karim

Plan due: October 14, 2026, 11:59 p.m.

Project Objective and Scope

Objective. Determine whether adapting the number of outstanding background reads (q) gives a better balance between foreground read latency and background scan throughput than any single fixed depth on a physical HDD. We will build one read-only user-space dispatcher and compare fixed depths $q \in \{1, 2, 4, 8\}$ against one feedback-based adaptive policy.

Scope. Linux; the physical HDD on the course server provided by the instructor; 4 KiB random foreground reads; fixed-size sequential background reads (256 KiB, with a second size for the robustness study); asynchronous I/O (io_uring/liburing) with direct I/O. Foreground admission limits and the installed Linux I/O scheduler stay identical across all policy comparisons. A local virtual disk or SSD may be used only for debugging; all reported performance results come from the server HDD.

Out of scope. Kernel scheduler changes, drive-geometry inference, writes to the HDD under test (other than creating approved test files), dynamic request sizes, and multiple background scanners.

Roles and Responsibilities

Member	Technical responsibility	Coordination responsibility
Mostafa Babaei	Common dispatcher, async I/O, correctness tests, optional RL extension.	Combine slides; maintain the code repository and reproduction instructions.
Muhammad Ahmad Tariq	Fixed and adaptive policies, controller settings, controller tests, analysis scripts.	Combine recordings; check playback; upload videos.
Sheryar Karim	Server/HDD setup, workload generator, experiment automation, measurements, plots.	Assemble the execution plan and final report.

Each member writes their own report section, prepares and presents their own slides, and records their own demonstration. The whole team reviews the design and every submission together before upload.

Measurements

Foreground p50/p99 latency, target-miss fraction, scanner MiB/s, completed scan bytes, configured and actual outstanding depth over time, pending queue lengths, rejected requests, controller CPU cost, and errors. Arrival, submission, and completion times are logged separately so waiting inside the program can be distinguished from submission-to-completion time (which includes software and device queueing and is not pure seek time).

Risks and Mitigation

- **Shared server HDD:** other users can distort latency. We will agree on time slots, record server load, and repeat runs.
- **Dispatcher bugs delay experiments:** workload generation, analysis scripts, and controller logic start early using sample logs.
- **Missing libraries on the server:** libaio or a thread pool is a fallback if liburing is unavailable.
- **Optional RL work:** done only after required work is complete, so it cannot delay any milestone.

Reproducibility and Correctness Rules

- Same arrivals, CPU allocation, file placement, scanner start point, warm-up, and cache preparation for every compared policy.
- Tuning runs are kept separate from final evaluation runs.
- Every accepted read completes exactly once with verified bytes or an explicit error; reads already submitted are never treated as canceled.
- Only approved disposable test files are used; no writes to any operating-system or personal-data drive.

Milestone 1: Environment Check and Design

Deadline: October 18, 2026, 11:59 p.m.
Goal: Confirm that the server HDD is usable, agree on the system design, and run a first smoke test. This milestone comes only four days after the plan, so it is kept small and focused on setup and design.

Note: The completion of the first milestone is contingent upon obtaining the required access to the course server. If this access is not granted in time, the corresponding tasks will be addressed in the second milestone.

Planned task	Responsible member
Log in to the course server; identify the HDD (rotational flag), record its model, interface, capacity, kernel version, and active I/O scheduler.	Sheryar Karim
Confirm the approved test directory, file-size limit, and whether other students share the HDD (and agree on testing time slots if so).	Sheryar Karim
Create the first test files with a known, verifiable data pattern.	Sheryar Karim
Check that gcc, liburing (or libaio), and Python plotting tools are available; request missing packages from the instructor.	Mostafa Babaei
Write a minimal smoke test: read the test file with direct I/O through io_uring and verify the bytes.	Mostafa Babaei
Design the dispatcher: foreground/background pending queues, request IDs, arrival/submission/completion timestamps, completion accounting.	Mostafa Babaei
Specify fixed depths $\{1,2,4,8\}$ and a draft adaptive rule (window size, minimum samples, miss threshold, step-down/step-up conditions).	Muhammad Ahmad Tariq
Define the log format (fields and units) shared by dispatcher, controller, and analysis scripts.	Muhammad Ahmad Tariq
Check video/slide requirements, upload platform, and deadlines.	Muhammad Ahmad Tariq
Write the first progress update; each member prepares and presents their own design slides.	All (assembled by Sheryar Karim)

Brief overview. We first verify that the server HDD, test files, and tools work, because every later measure-

ment depends on them. In parallel we fix the dispatcher design, the log format, and a draft feedback rule so coding can start immediately after this milestone.

Milestone 2: Working Dispatcher and Fixed-Depth Baselines

Deadline: November 1, 2026, 11:59 p.m.
Goal: Build a correct dispatcher and measure the foreground-latency versus scan-throughput tradeoff for every fixed depth (Study A).

Planned task	Responsible member
Implement the common dispatcher: asynchronous reads, two pending queues, foreground quota, background limit q, timestamps, completion accounting, clean shutdown.	Mostafa Babaei
Correctness tests: verified file contents, end-of-file, full pending queue, read errors, shutdown with reads in flight.	Mostafa Babaei
Implement configurable fixed depths $q = 1, 2, 4, 8$ and record the *actual* outstanding count, not only the configured cap.	Muhammad Ahmad Tariq
Write analysis scripts: p50/p99 latency, target-miss fraction, MiB/s, completed scan bytes.	Muhammad Ahmad Tariq
Build the workload generator (fixed foreground arrival schedule, recorded rejected requests) and run scripts with consistent warm-up and cache preparation.	Sheryar Karim
Run foreground-alone, scanner-alone, and combined runs at each fixed q (3 repetitions); plot foreground p99 versus scan throughput.	Sheryar Karim
Use these results to choose and *freeze* the latency target and controller thresholds.	Muhammad Ahmad Tariq with team
Prepare slides and recordings for the dispatcher/tests, fixed policies, and baseline results (one part per member).	All

Brief overview. We implement one shared dispatcher so every policy pays the same overhead, test its invariants, and then measure how foreground p99 latency and scan throughput change with each fixed depth. These baseline results set the frozen latency target used by the adaptive controller.

Milestone 3: Adaptive Policy and Controlled Experiments

Deadline: November 15, 2026, 11:59 p.m.
Goal: Implement the feedback controller and compare it with all fixed depths under constant and changing demand, plus the boundary and robustness studies (Studies B and C).

Planned task	Responsible member
Implement the windowed target-miss controller with minimum sample count, hold-on-sparse-data, $q \geq 1$ while scanning, and thresholds in a config file.	Muhammad Ahmad Tariq
Implement the no-damping ablation (no multi-window increase rule) and unit tests for sparse windows and allowed q values.	Muhammad Ahmad Tariq
Verify safe limit reductions: when q drops below the outstanding count, stop new background submissions until in-flight reads finish.	Mostafa Babaei
Create the constant-load and alternating quiet/busy traces with identical offered requests for every policy.	Sheryar Karim
Run Study B (adaptive vs. all fixed depths, 3 repetitions) and plot q, target misses, and cumulative scan bytes over time.	Sheryar Karim
Run Study C: one demanding overload case, and the changing-load comparison with a second chunk size (e.g., 1 MiB).	Sheryar Karim and Mostafa Babaei
Check data integrity, errors, and completion accounting on every run; measure controller CPU cost.	Mostafa Babaei
Diagnose oscillation and slow recovery; compare with the ablation.	Muhammad Ahmad Tariq
Optional, only if required work is on schedule: prototype an RL policy choosing $q \in \{1, 2, 4, 8\}$, trained on separate tuning traces.	Mostafa Babaei
Prepare slides and recordings for the controller, experiments, and correctness checks (one part per member).	All

Brief overview. With the target frozen, we run the adaptive controller and all four fixed depths on identical traces and repeat each principal comparison at least three times. The ablation, overload case, and second chunk size show whether any benefit is real and robust rather than an artifact of one setting.

Milestone 4: Analysis, Report, Presentation, and Submission

Deadline: December 1, 2026, 11:59 p.m.
Goal: Explain the findings and submit the complete code, data, report, slides, and videos.

Planned task	Responsible member
Produce final latency-throughput and time-series figures; summarize repeated runs with variability.	Sheryar Karim
Analyze controller behavior, oscillation, recovery, and target misses; write the controller section.	Muhammad Ahmad Tariq
Write the implementation and testing section, including I/O-path limitations (application queue depth $\neq$ firmware queue depth; drive cache remains).	Mostafa Babaei
Write the setup and results section (hardware, scheduler, cache preparation, request counts).	Sheryar Karim
If the RL prototype is viable, evaluate its frozen policy on held-out traces and report benefit, training cost, and overhead.	Mostafa Babaei
Assemble the 4 to 6 page IEEE/ACM conference-format report.	Sheryar Karim
Organize the repository: code, configs, tests, workloads, raw logs, plots, reproduction instructions.	Mostafa Babaei
Combine slides; combine and upload recordings; verify access.	Mostafa Babaei / Muhammad Ahmad Tariq
Rehearse and present each member's part.	All

Brief overview. We turn the logs into the required figures and state clearly whether adaptation beats the best fixed depth; a result showing a fixed depth is just as good is reported as a valid finding. Each member completes their section, slides, and video, and the team reviews everything before submission.